Signed power recovery and small-coefficient quartics for the consistency of ZFC

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with AI assistance and independent computational review

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Abstract

This investigation produces two concrete improvements in the direct proof-certificate branch of the ProveIt consistency project. A polynomial radius in nine Euclidean quotients makes the associated terminal power outputs admissible before their signs are known. Their power equations then recover the omitted signs. After exact integer elimination this gives a quartic in **577 unrestricted integer variables**, with **7,326 monomials** and coefficient height $2,445,790,974,704$. The matched predecessor has 604 variables and 9,676 monomials at the same degree and height. A lower-height companion has 582 variables, 6,069 monomials and height 964,213.

A separate construction replaces repeated coefficients and linear occurrences by private variables forced to equal one. Applied to the squared-output unprojected system, it yields a quartic in **3,564 unrestricted integer variables** whose nonzero coefficients are exactly drawn from $\{-2, 1, 2\}$. Its 14,518 coefficients are supplied explicitly and checked in full.

Both constructions preserve the existence of a finite syntactic ZFC refutation, using the inherited numerical C222 calibration. Consequently their unsolvability is provably equivalent in ZFC to Con(ZFC), by ordinary mathematical arguments formalizable in ZFC. Neither construction uses K215 or a Jones universal polynomial. The new arguments are not presented as completed Lean or object-language ZFC formalizations. No global minimum, historical novelty theorem, consistency proof, or improvement over the repository's fewest-variable branch is claimed.

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1 Results and the meaning of simplicity

1.1 The explicit equations

Let $r_0, \dots, r_{362} \in \mathbb{Z}[x_0, \dots, x_{576}]$ be the quadratic polynomials printed in Appendix A. Define

$$\boxed{P_{577}(x_0, \dots, x_{576}) = \sum_{j=0}^{362} r_j(x_0, \dots, x_{576})^2 = 0, \qquad x_0, \dots, x_{576} \in \mathbb{Z}.} \tag{1}$$

This is one ordinary polynomial equation with fixed integer coefficients. The sum is finite and completely specified. The subscripts of its coordinates are labels, not parameters. In particular, it contains no variable exponent, hidden bounded quantifier, unspecified universal-polynomial parameter, or coefficient selected according to whether ZFC is consistent.

The machine-readable quadratic rows are in `quotient_frame/support8/equation.json`. The fully collected polynomial, with all 7,326 coefficients, is in the adjacent `polynomial.json`. The archive also supplies the complete lower-height equation $P_{582} = 0$ in `quotient_frame/support16_affine2/`. It is the sum of the squares of 368 explicitly listed quadratic rows.

The second main construction is

$$\boxed{Q_{3564}(z_0, \dots, z_{3563}) = \sum_{j=0}^{3349} s_j(z_0, \dots, z_{3563})^2 = 0, \qquad z_0, \dots, z_{3563} \in \mathbb{Z}.} \tag{2}$$

Its complete rows, coefficient list and reconstruction certificate are in `height_two/anchored/`. Section 3 gives a finite construction of every row from the supplied 1,457-coordinate parent. The large subscript counts actual unknowns; it does not hide fixed coefficient payloads.

Theorem 1.1 (Consistency endpoint)**.** *For each of the supplied polynomials $P_{577}, P_{582}, P_{586}, P_{591}, Q_{3564}$, ZFC proves*

$$(\textit{there is no zero over the stated power of } \mathbb{Z}) \quad \longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

The consistency predicate is the absence of a finite syntactic refutation in the fixed ZFC calculus used by the inherited source calibration. The proof is an ordinary mathematical proof using that identified calibration, together with the new reductions proved below; the new entire equivalence has not been checked in a proof assistant.

The construction proves an equivalence, not either side of that equivalence. A finite refutation supplies finite integer witnesses by the completeness construction. Conversely, any integer zero can be reconstructed into a finite refutation. These are unbounded statements: no bound is placed on proof length, Pell indices, or the magnitudes of the unknowns.

1.2 Measured improvements

We use *coefficient height* to mean the maximum absolute value of a coefficient of the fully collected polynomial. The count of quadratic residuals refers to its displayed sum-of-squares representation; it is not the number of equations in the final problem, which is one.

The quotient-radius change removes 27 final unknowns and nine quadratic residuals. The 577-variable equation has 2,350 fewer monomials than its matched 604-variable predecessor, a 24.29% reduction. The lower-height 582-variable equation has 2,754 fewer monomials than its matched 609-variable predecessor, a 31.21% reduction, with the same height 964,213.

Construction	Variables	Rows	Monomials	Height
Matched direct predecessor	604	372	9,676	2,445,790,974,704
New quotient-radius equation	577	363	7,326	2,445,790,974,704
Squared-output companion	586	372	7,059	2,445,790,974,704
Matched lower-height predecessor	609	377	8,823	964,213
New quotient-radius companion	582	368	6,069	964,213
Squared-output, lower height	591	377	5,835	964,213
New coefficient-two equation	3,564	3,350	14,518	2

Table 1: All rows describe unrestricted integer unknowns and exact degree four. Baselines are fixed at the inspected repository snapshot.

The squared-output radius, developed first, supplies another useful tradeoff: its 586-variable form has 7,059 terms, and its 591-variable lower-height form has 5,835 terms. Thus these forms have nine more variables but fewer terms than the corresponding quotient-radius forms. All four are retained. The lower-height variants have five more variables and substantially smaller coefficients than their corresponding high-coefficient variants.

At source level nine nonnegativity conditions disappear, saving 27 private sign witnesses. Squaring the terminal outputs prevents nine old coordinate eliminations, leaving a net saving of 18. Squaring the quotient coordinates instead restores those eliminations and realizes the full saving of 27. All counts are measured from complete polynomial artifacts.

For Q_{3564}, the full coefficient histogram is

coefficient	-2	1	2
number of monomials	$4,470$	$7,280$	$2,768.$

In particular, no large integer has merely been moved to an unexplained coefficient port. The logical source is encoded in the finite arrangement of elementary rows.

1.3 What is and is not improved

There is no established single ordering called “simplest Diophantine equation.” The useful objectives here include number of variables, total degree, coefficient height, number of monomials, arithmetic circuit size, and the length of the complete fixed-data recipe. These objectives conflict.

The pinned repository already contains nine-positive-variable and ten-unrestricted-integer constructions, as well as 38-unrestricted-integer quartics obtained through the native coefficient route. The present 577-variable equation does not improve those variable counts. Its improvement concerns the direct C222 proof-certificate quartics with moderate literal coefficients. The coefficient-two equation is a further tradeoff in the coefficient-height direction. It is not evidence that 3,564 is the fewest number of variables possible at height two.

The contribution is therefore two specific constructions and two reusable mechanisms. The first replaces a sign-dependent magnitude bound by a sign-independent one and then recovers signs from arithmetic. The second prevents accumulation of coefficients by separating occurrences with private unit variables. The investigation does not establish that either general device is historically new throughout the literature. The changes are new relative to the inspected construction, and the supplied numerical gains are independently checkable.

1.4 Frozen comparison point

All source comparisons use ProveIt commit

`84e16e43092614bae262ca59651cb9d77e31a255`.

The collection README is older than some active research files. The actual baselines come from `research/factorial-radius-output-sign-profile/`, whose 75-predicate family includes the 604- and 609-variable equations above. We also inspected the active size comparison and the newer incoming archives before selecting this branch. No earlier repository construction is removed or replaced by this package.

2 Recovering signs from bounded power certificates

2.1 Why the old order of reasoning costs variables

The direct arithmetic source has a finite collection of proposed power identities $m = n^k$. A common Pell frame provides a parameter large enough to certify all these powers simultaneously. Its former radius was the sum of the distinct power bases, exponents and proposed outputs, plus two. This is useful when those quantities are already known nonnegative.

Several proposed outputs obtain that sign only from Euclidean-division rows. For example, an output may satisfy

$$m = MLq + r, \qquad L = r + s + 1.$$

If $M \geq 1$ and $q, r, s \geq 0$, then $m \geq 0$. The source therefore paid for a nonnegativity certificate for q, so that m could occur in the radius, so that the power certificate could prove $m = n^k$. Once the power is known, $m > 0$ already forces $q \geq 0$. The apparent circularity can be removed by changing how the radius bounds m.

Let $\mathcal{A}$ be the 43 distinct original frame atoms. Let $\mathcal{M} \subset \mathcal{A}$ be the nine selected power outputs. Replace the final frame-definition row by

$$\boxed{W = 2 + \sum_{a \in \mathcal{A} \setminus \mathcal{M}} a + \sum_{m \in \mathcal{M}} m^2.} \tag{3}$$

The 34 unselected atoms retain direct or derived nonnegative signs. The selected nine are allowed to be arbitrary integers at this stage. Nevertheless,

$$0 \leq a \leq W - 2 \quad (a \notin \mathcal{M}), \qquad m^2 \leq W - 2 \quad (m \in \mathcal{M}),$$

so every selected output satisfies $|m| < W$. Squaring the outputs changes the frame row from linear to quadratic; its square remains quartic in the final equation.

2.2 An exact signed-output lemma

For an integer $A \geq 2$, define the Pell sequences by

$$\chi_A(j) + \psi_A(j)\sqrt{A^2 - 1} = \left(A + \sqrt{A^2 - 1}\right)^j \qquad (j \geq 0).$$

These expressions abbreviate the integer sequences given by the initial values $\chi_A(0) = 1, \psi_A(0) = 0$, $\chi_A(1) = A, \psi_A(1) = 1$, and recurrence $u_{j+2} = 2Au_{j+1} - u_j$.

Lemma 2.1 (Signed power recovery)**.** *Suppose W, A, n, k, m are integers satisfying*

$$W \geq 4, \quad A > W^W, \quad 2 \leq n < W, \quad 1 \leq k < W, \quad |m| < W.$$

Put $T = 2An - n^2 - 1$. Suppose also

$$Y = \psi_A(k), \qquad X \in \{\chi_A(k), -\chi_A(k)\}, \qquad T \mid X - (A-n)Y - m.$$

Then $X = \chi_A(k)$ and $m = n^k$.

Proof. First $T > A$. Indeed $n \geq 2$, $n \leq W - 1$, and $A > W^W$ give

$$T - A = (2n-1)A - n^2 - 1 > A - (W-1)^2 - 1 > 0.$$

Set $x = \chi_A(k)$, $y = \psi_A(k)$, $p = x - (A-n)y$, and $q = x + (A-n)y$. The Pell norm gives the exact identity

$$pq = x^2 - (A-n)^2 y^2 = 1 + Ty^2.$$

The recurrence gives $p \equiv n^k \pmod{T}$: its initial values are 1 and n, and modulo T the sequence n^j satisfies the same recurrence, because $n^2 - 2An + 1 = -T$.

If $X = -x$, the proposed power row gives $m \equiv -q \pmod{T}$. Consequently $T \mid mn^k + 1$. This multiple is nonzero: $n^k \geq 2$ cannot divide -1. On the other hand,

$$|mn^k + 1| \leq (W-1)^W + 1 < W^W < T,$$

a contradiction. The strict inequality follows directly by expanding $((W-1)+1)^W$.

Thus $X = x$. The row now gives $T \mid m - n^k$, while

$$|m - n^k| \leq (W-1) + (W-1)^{W-1} < W^W < T.$$

The multiple must be zero, proving $m = n^k$. □

The lower bound $n \geq 2$ is essential. If $n = 1$, the negative-root branch admits $m = -1$, and $mn^k + 1 = 0$ defeats the argument. Every selected call below has the same source base, raw coordinate 531, whose retained constructor proof gives a value at least two. The transformation is not applied to arbitrary base-one calls.

2.3 The nine actual deletions

Raw indices refer to the frozen 1,259-coordinate body before projection. They are zero-based labels, not additional unknowns. The omitted scalar guards and their associated calls are as follows.

Call	Output	Exponent	Retained index slack	Omitted guard
7	546	110	825	548: exact-binomial high quotient
10	573	44	891	575: range quotient
12	583	46	934	584: range quotient
13	591	48	964	592: range quotient
14	599	50	994	600: range quotient
15	607	52	1024	609: range quotient
17	617	54	1084	618: range quotient
18	625	56	1114	627: range quotient
20	635	58	1174	636: range quotient

Precisely source row 1176 is replaced by (3). All other original body rows and variable records are retained. The former intermediate summation gates may still define unused bookkeeping coordinates; the integer projection eliminates such coordinates through their defining rows. No unconstrained variable is silently deleted.

The 75-predicate parent becomes a 66-predicate parent. A nonnegative expression E is encoded by

$$E = a^2 + b^2 + c(c+1), \qquad a, b, c \in \mathbb{Z}. \tag{4}$$

The right side is always nonnegative for integer inputs. Conversely every nonnegative integer is represented: apply the three-square theorem to $4E+1$, whose representation has one odd and two even coordinates. Dividing by four gives (4). Thus these rows are exact sign conditions, with no implicit natural-number domain on the witnesses.

The new unprojected system has $1259 + 3 \cdot 66 = 1457$ variables and $1177 + 66 = 1243$ quadratic rows. In particular, all sign requirements used in the proof are actual polynomial constraints.

2.4 The bootstrap must precede the power interpretation

Deleting the quotient guards also removes some old intermediate sign deductions. Reusing the entire old sign ledger without checking its premises would be invalid. The new certificate instead filters it in dependency order.

One initial sign is recovered before this filtering. Let f, N, r, s, C denote raw coordinates $87, 93, 94, 95, 91$. The unchanged rows give

$$C = r + s + 1, \qquad fC = N + s,$$

and hence

$$N = fr + (f-1)s + f \geq 2,$$

using only the retained conditions $f \geq 2$, $r, s \geq 0$. No factorial or power interpretation is used here.

With that initial fact and the retained scalar predicates, 605 of the 643 original prefix-coordinate signs are certified. The 38 missing prefix signs lie in the nine deleted quotient/output blocks and their local arithmetic. Every unselected frame atom still has a nonnegative sign. All 431 originally listed strict-positive constructor deductions remain licensed. The selected exponents are positive; their actual retained index-slack constraints give $Y \geq k \geq 1$.

The unchanged initial modulus guard establishes $A \geq 2$. The bootstrap Pell row has the form

$$A^2 - W^3(W+2)Z^2 = 1.$$

The standard classification at base $W+1$ writes $W|Z| = \psi_{W+1}(\ell)$. Since $A \geq 2$, the bootstrap row forces $Z \neq 0$. The congruence $\psi_{W+1}(\ell) \equiv \ell \pmod{W}$ therefore makes ℓ a positive multiple of W. No sign of the original Z is assumed. Therefore

$$A \geq \chi_{W+1}(W) > W^W.$$

The magnitude conclusions of (3) supply all the bounds required by Lemma 2.1.

The common unit and the signed comparison-base argument are unchanged. They use the retained strict common-product guard, the exact unit norm, and each selected index slack to recover

$$Y = \psi_A(k), \qquad |X| = \chi_A(k).$$

They do not require the omitted quotient signs or prior power meanings. The original polynomial row identities supply the required divisibility relation. Thus Lemma 2.1 proves all nine selected outputs positive with their exact power values.

2.5 Restoring the omitted inequalities

For a range quotient, the unchanged exact identities are

$$m = MLq + r, \qquad L = r + s + 1,$$

where $M \geq 1$, $r, s \geq 0$, so $L \geq 1$ and $0 \leq r < L$. If $q \leq -1$, then

$$m \leq -ML + r \leq -L + r < 0,$$

contrary to the recovered value $m = n^k > 0$. Hence all eight omitted range quotients are nonnegative.

The extra exact-binomial quotient has the identities

$$m = L(C + bq) + r, \qquad L = r + s + 1, \qquad b = C + z + 1,$$

where $C, r, s, z \geq 0$. If $q \leq -1$, then $C + bq \leq C - b \leq -1$, again giving $m \leq -L + r < 0$. This proves raw coordinate 548 nonnegative. The other two exact-binomial high-quotient guards are retained: their seed-index dependencies do not license the same early use of this lemma.

After these nine signs have been recovered, all premises of the old prefix and factorial/index arguments are restored. The remaining proof can establish the exact arithmetic meanings in its original order: the powers, the factorial/binomial identities, the CRT range conditions, and finally the finite proof-certificate condition.

2.6 Soundness and completeness are existence maps

The new radius changes the private Pell data. It is false that an arbitrary old integer zero becomes a new zero merely by deleting guard witnesses. Nor does a new zero satisfy the old final frame-definition row at the same coordinates.

For soundness, first apply the preceding bootstrap and signed-power recovery at the new source tuple. The old arithmetic proof uses the frame-definition equality to obtain magnitude bounds and a large Pell parameter; those conclusions have just been established directly for the new radius. It does not use the literal sum elsewhere in the semantic computation. The remaining original source proof therefore produces the same finite C222 derivation and syntactic refutation. Equivalently, once the genuine semantic data are known, the old private frame can be rebuilt. The certificate records that the original power arguments are allocated before the private Pell coordinates, so that this rebuilding does not change those arguments.

For completeness, begin with a genuine finite derivation and its original semantic arithmetic witnesses. Compute the true terminal powers, then define the new W by (3). Take

$$A = \chi_{W+1}(W), \qquad Z = \psi_{W+1}(W)/W.$$

The quotient is an integer by the displayed index congruence. Construct the main pairs at their fixed indices, then choose a common unit and comparison parameter by the inherited finite-period and CRT construction. Finally construct the progression quotients and the ordinary power quotients. Every requested base, exponent and output was fixed before these private parameters. This is a finite ordered construction, not a simultaneous fixed-point assumption.

All retained guard values have their required signs in this canonical construction. Choose fresh triples for (4). This gives a zero of the complete new 1,457-variable system. The exact integer projections then give zeros of P_{586} and P_{591}.

Proposition 2.2. *The new 66-predicate integer source has a zero if and only if the calibrated finite C222 proof certificate exists. Its two projected quartics have integer-zero existence equivalent to that source.*

Proof. The preceding two constructions give the source equivalence. For projection, each eliminated coordinate has a unique integer polynomial restoration obtained from an actual monic defining row. Substitute it into every surviving row, including the guard rows. Reverse these substitutions to lift any projected zero. The complete identities and variable partitions are supplied and independently checked. No sign assumption is used during this algebraic lifting; signs are interpreted only at the restored source tuple. □

2.7 A stronger radius in the quotients themselves

The preceding construction bounds each selected output directly by its square. It can be improved by bounding the quotient that produces that output. Let $\mathcal{Q}$ be the nine omitted quotient coordinates listed above. Use instead

$$\boxed{W = 5 + \sum_{a \in \mathcal{A} \setminus \mathcal{M}} a + \sum_{q \in \mathcal{Q}} q^2.} \tag{5}$$

The nine terminal outputs are absent from this expression. All other source rows and all 66 predicates are the same as before. This is the source for the preferred 577-variable equation.

Every unsigned atom now satisfies $0 \le a \le W - 5$, and each omitted quotient satisfies $|q| < W$. In particular, every selected base and exponent obeys

$$2 \le n \le W - 5, \qquad 1 \le k \le W - 5,$$

so $W \ge 7$. The same independent prefix reasoning is available.

Write b for the radix, raw coordinate 530, and M for raw coordinate 129. The unchanged constructor rows give

$$b = M(1 + N + N'), \qquad M \ge 1, \quad N \ge 2, \quad N' \ge 0.$$

Thus $1 \le M \le b < W$ before any intended product or factorial interpretation. Each range-block denominator L is an unsigned frame atom, so $L < W$. Its remainder row gives $0 \le r < L$. The exact identity $m = MLq + r$ consequently implies

$$|m| \le ML|q| + r < W^3 + W < W^4.$$

For the extra binomial quotient, $b = C + z + 1$ gives $0 \le C < b < W$, and

$$|m| \le L(C + b|q|) + r < W^3 + W^2 + W < W^4.$$

These bounds use neither the omitted signs nor a selected power's meaning.

Lemma 2.3 (Exponent margin compensates for a polynomial output bound)**.** *The conclusion of Lemma 2.1 still holds if its bounds are replaced by*

$$W \ge 7, \quad A > W^W, \quad 2 \le n \le W - 5, \quad 1 \le k \le W - 5, \quad |m| < W^4.$$

Proof. Again $T > A$. The negative-root case would make the nonzero integer $mn^k + 1$ a multiple of T. But

$$|mn^k + 1| \le (W^4 - 1)W^{W-5} + 1 \le W^{W-1} < W^W < T.$$

For the positive-root case,

$$|m - n^k| < W^4 + W^{W-5} \le 2W^{W-1} < W^W < T.$$

The same congruence arguments therefore recover $X > 0$ and $m = n^k$. □

This is the main strengthening. A polynomial magnitude bound on a proposed output suffices when the exponent has a fixed margin below the common radius. The radius constant five supplies that margin without an extra variable.

2.8 Selective reuse of the inherited power proof

Equation (5) does not imply that every original proposed output is smaller than W, and it is not asserted to be larger than the old linear radius. The proof must account for this explicitly.

The nine selected calls are settled by Lemma 2.3. Their positive values then restore the nine omitted quotient signs by the same division arguments as before. The other twelve power calls retain their direct output bounds from the 34 unsigned frame atoms. In particular, the inherited local factorial and missing-index bootstrap interprets calls 1, 5 and 6 and orients calls 0 and 9; none is among the selected nine. Its parity step also uses the unchanged row for call 2, without presupposing that power's meaning. Call 2 is also among the twelve retained calls. Every magnitude bound needed in that bootstrap is therefore still available. One replays its argument for those twelve calls, using the already proved selected powers when needed. This establishes all 21 requested powers without applying an invalid global bound to the selected outputs.

Once the powers are known, the later factorial, binomial, CRT and proof-decoding arguments no longer need a bound $m < W$ on a selected terminal output. Soundness reaches the same finite refutation.

For completeness, choose genuine semantic data and true powers first. The nine quotients are then ordinary Euclidean witnesses fixed by those data. Define W by (5), and rebuild all private Pell parameters in the previous order. Every base and exponent is smaller than W, so its true power is smaller than $W^W < A$; this supplies the size condition needed by the canonical power-witness construction even when that output is larger than W. There is no dependence of a radius input on a subsequently chosen private Pell variable.

Corollary 2.4. *The quotient-radius 66-predicate source has a zero exactly when the calibrated finite C222 proof certificate exists. Exact integer elimination gives P_{577} with 577 variables and 363 rows, and P_{582} with 582 variables and 368 rows. Both have exact degree four.*

The quotient-radius system removes all 27 private sign witnesses and leaves the old terminal-output eliminations available. Its 577-variable projection has 880 explicit integer restorations; the 582-variable projection has 875. These are full inverse certificates against the actual source, not dimension estimates.

3 A quartic whose coefficients have absolute value at most two

3.1 A different objective

The quotient-radius equations reduce the number of variables in the direct family. They still have coefficients inherited from elimination. A separate construction gives a particularly simple coefficient alphabet by retaining more intermediate variables.

For this purpose the slightly smaller input is the squared-output 66-predicate parent, not the quotient-radius parent. Both have 1,457 coordinates and 1,243 rows; the latter's different frame row produces a slightly larger unit construction. We retain the better coefficient-height point, with 3,564 variables, rather than force all improvements into a single branch.

Write the old coordinates as $x_0, \dots, x_{1456}$, add product coordinates $p_0, \dots, p_{164}$, and add unit coordinates

$u_0, \dots, u_{1941}$. Every coordinate ranges over $\mathbb{Z}$. Define

$$U_0 = u_0 - 1, \qquad U_j = u_j - u_{j-1}^2 \quad (1 \le j \le 1941),$$

and

$$T_h = p_h - x_{a_h} x_{b_h} \quad (0 \le h \le 164).$$

The 165 pairs (a_h, b_h) are fixed entries of the supplied certificate. Let R_i, $0 \le i \le 1242$, be the transformed source rows described below and supplied in full. The equation is

$$\boxed{Q_{3564} = U_0^2 + \sum_{j=1}^{1941} U_j^2 + \sum_{h=0}^{164} T_h^2 + \sum_{i=0}^{1242} R_i^2 = 0.} \tag{6}$$

These are fixed finite sums. In particular the unit recurrence is a fixed list of quadratic equations, not a variable-length condition.

The variable count is

$$1457 + 165 + 1942 = 3564,$$

and the residual count is

$$1243 + 165 + 1942 = 3350.$$

The normative expanded polynomial has 14,518 monomials and constant coefficient one.

3.2 Why small row coefficients do not suffice

A residual such as $z - x - y$ has coefficients only $1, -1$. But a sum of many such residual squares can have a large coefficient of x^2, because every use of x contributes another copy. Small local coefficients therefore do not imply a small height for the collected final polynomial.

Assign distinct private units to distinct linear occurrences. For example, replace

$$z - x - y \quad \text{by} \quad z - xu_i - yu_j.$$

At a zero of (6), the unit rows force all $u_j = 1$, so the intended equation is unchanged. Its square, however, contains $x^2 u_i^2$ and $y^2 u_j^2$. These are distinct monomials from those produced by other uses of x or y.

A larger coefficient is split into separate marked occurrences. For example,

$$-4x \quad \text{becomes} \quad -xu_a - xu_b - xu_c - xu_d,$$

with four distinct private units. A constant occurrence is represented by a private unit alone. Thus a constant -2 becomes $-u_e - u_f$. This pays explicitly for each occurrence and prevents coefficient accumulation after collection.

An original triangular gate output may remain unmarked in its unique defining row. A further linear term can remain unmarked when its variable has no other unmarked linear occurrence. If its row also retains a pure quadratic monomial xy, the unmarked coordinate z is required to have an index larger than both product inputs. This makes z distinguishable in the resulting cubic cross term. There are 152 additional safe unmarked occurrences in the preferred construction, saving 152 units and residuals relative to its simpler 3,716-variable predecessor.

3.3 Keeping degree four

Multiplying a quadratic occurrence xy by a unit would make a cubic residual and could raise the final degree to six. Instead, reuse a product coordinate or introduce one. If an original row is literally $z - xy = 0$, it can supply the product coordinate. There are 383 distinct canonical original product definitions in this instance. For any other product that must be replaced, introduce one shared row $p - xy = 0$; 165 new product coordinates suffice.

Every transformed source row retains at most one pure quadratic monomial. The construction ensures that a given raw quadratic monomial is retained at most twice across all transformed and new-product rows. Other occurrences use its product coordinate, with a private unit where required. Every row still has degree at most two.

This sharing requires a precise proof. An old product coordinate does not equal its intended product at an arbitrary off-zero tuple. Its identity is available because the literal defining row is retained. For each transformed source row the independent verifier checks an identity of the form

$$R_i\big|_{u=1,\ p=\text{products}} = A_i + \sum_{(x_a x_b \rightsquigarrow z)} c_{i,a,b}(z - x_a x_b), \tag{7}$$

where A_i is the original residual and every correction term is a genuine canonical original product definition. No pointwise substitution of an unconstrained old wire is assumed.

3.4 Complete coefficient collision argument

The following elementary conditions are checked for the entire instance: each private unit belongs to exactly one source term; each unmarked linear coordinate occurs at most once; each row has at most one unmarked linear term and at most one pure quadratic; every such raw product occurs at most twice; and every unmarked linear/pure-product cross term has its linear coordinate strictly larger than the product inputs.

Proposition 3.1. *Under those conditions, the coefficient alphabet of* (6) *is contained in* $\{-2, 0, 1, 2\}$.

Proof. The signal coordinates and unit coordinates have disjoint index ranges. Consider the possible monomials after squaring.

The unit seed contributes $u_0^2 - 2u_0 + 1$. A later unit row contributes $u_j^2 + u_{j-1}^4 - 2u_j u_{j-1}^2$. These monomials do not coincide with one another. A private constant occurrence may contribute one further u_j^2, giving coefficient two. Each unit has at most one such occurrence. The seed unit has no source occurrence.

An unmarked linear signal contributes its square once. A retained pure quadratic xy contributes x^2y^2 once or twice, according to its global multiplicity. There is no cross product between two pure quadratics in one row. Thus the purely signal quartics have coefficients one or two.

The purely signal cubic cross terms have the form $\pm 2zxy$, with $z > \max(x, y)$. The largest index identifies the unique unmarked linear occurrence, so different rows cannot produce the same such cross term. New-product outputs satisfy the same property because they follow all source coordinates.

A squared marked term has a squared private-unit factor, distinguishing it from cross terms. A cross term involving two marked occurrences is distinguished by its pair of private unit indices. A cross term with one marked occurrence is distinguished by its unit, and its other factor is the unique unmarked linear term or unique retained pure quadratic in that row. The different degrees distinguish these cases.

If two marked occurrences are constants, their cross term has two distinct units; it cannot equal a unit-chain square or cubic. All cross-term coefficients are 2 or -2.

These cases exhaust every term in every residual square. No coefficient can accumulate to absolute value greater than two. □

The supplied equation attains each of $-2, 1, 2$. Its coefficient of u_0^4 is exactly one, from U_1^2, so its degree is exactly four. Only U_0 has a literal constant term, so the polynomial's constant coefficient is exactly one. The complete independently collected histogram is

$$\#(-2) = 4470, \qquad \#(1) = 7280, \qquad \#(2) = 2768.$$

3.5 A genuine bijection of zero sets

Theorem 3.2. *Coordinate projection gives a bijection from the integer zero set of Q_{3564} to the integer zero set of its supplied 1,457-variable squared-output source. Its inverse is a fixed polynomial map: set all units to one and set each new product coordinate to its designated product of source coordinates.*

Proof. At an original source zero, the canonical old product rows give their intended products. The proposed extension makes every unit and new-product residual zero. Identity (7) then makes each transformed source residual zero.

Conversely, a zero of the integer sum of squares makes every residual zero. The seed and unit chain force all units to one. The new product rows fix all new product coordinates. The canonical original product rows have been preserved and fix their old product coordinates. Apply (7) to recover every original residual. Each additional coordinate is uniquely determined, proving bijectivity. □

Unlike the change of Pell radius, this coefficient transformation does give a polynomial bijection of zero sets. The distinction matters: the radius change rebuilds private number-theoretic witnesses, whereas the unit construction only attaches deterministic arithmetic coordinates.

3.6 A general construction beyond this instance

The private-unit construction works for any finite quadratic integer system. A simple version first introduces a product coordinate for every distinct quadratic monomial and then marks every original linear or constant occurrence. It needs no special triangular outputs. If the source has n variables, r rows, D distinct quadratic monomials and total coefficient absolute-value sum L, this version uses at most

$$n + D + L + 1 \quad \text{variables}, \qquad r + D + L + 1 \quad \text{quadratic residuals}.$$

Its final quartic has coefficients in $\{-2, 0, 1, 2\}$ and the same projected integer zero set. The delivered construction improves these coarse counts by reusing original product definitions and safe unmarked terms.

This general observation does not imply a universal small bound on the number of variables, because D and L depend on the source. Its usefulness here comes from the direct C222 arithmetic source having a moderate explicit row list and small local coefficients.

4 From the numerical source to consistency

4.1 The logical source is independent of K215

The starting point is the repository's numerical C222 calculus, used directly as a proof-certificate source. The name is historical; this construction does not encode the tape or transition table of a 222-state or 215-state machine. Nor does it substitute an unspecified program index into Jones's universal polynomial.

The inherited source has explicit arithmetic constructors for formula codes and a fixed table of rule alternatives. The projected certificate uses eight numerical columns: four rule/schema parameters, two consecutive accumulator values, and two beta-code quotient columns. The truth and unconditional axiom alternatives have no nontrivial premises. Modus ponens and generalization retain their actual premise and conclusion constructors. The guarded axiom alternative retains the arithmetic test that distinguishes its required parameters. The source archive contains the complete rule table and constructor data; the new radius transformation changes none of them.[1, 2]

The accumulator uses weights $w(j) = 1 + Mj$ and transitions

$$S_{i+1}w(v_i) = S_i w(p_i)w(q_i).$$

For a finite valid certificate these equations telescope to an equality of products of linear factors. A sufficiently large explicitly generated carrier allows that numerical equality to be decoded coefficient by coefficient into a polynomial identity. The inherited one-sided degree argument bounds the degrees on both sides before using base uniqueness; it does not assume the desired balance in advance.

The polynomial flow identity implies derivability of the target even if the proposed rule multiset contains cycles or repeated premises. One can evaluate a suitable linear functional supported outside the derivability closure: every permitted rule respects that closure, so an unmatched target outside it contradicts the flow identity. Completeness expands a finite derivation into a finite proof tree and pads it with truth steps. Thus this is an unbounded finite-derivation certificate, rather than a certificate for proofs below a fixed external bound.

4.2 Why the arithmetic compiler has finitely many variables

The eight columns are encoded by common CRT and beta representations. Exact factorial and binomial constructions supply the necessary moduli and low-digit extraction. The range certificates use binomial divisibility, which permits the high-digit quotients to be auxiliary values rather than independent semantic columns. All variable-length computation is represented by arithmetic identities with fixed finite witness lists.

The shared Pell construction certifies all 21 required powers with one common frame and a finite collection of seed-index tests. Its auxiliary units and congruences can be chosen simultaneously by finite period arguments. All power bases, exponents and proposed outputs are allocated before the private Pell coordinates. That separation is the reason the new frame can be rebuilt after its semantic inputs are fixed.

The compiler's numerical values can be enormous. Finiteness, exact integral identities, and an unbounded proof of correctness are what matter for the Diophantine representation. A practical search for a ZFC refutation, a feasible solution bound, and an actual standard integer zero are not claimed.

4.3 The exact calibration dependency

The preserved source archive contains the immediate Lean source files for the named endpoint theorems

`KEndpoint.c222_iff_zf_refutation`

and

`ConstructibleChoice.zf_refutation_iff_zfc_refutation`.

The first is unconditional and occurs before the later machine-parameter portion of its file. The second transfers syntactic refutation existence between ZF and ZFC using the inherited constructibility argument. Together their intended ordinary mathematical content is

$$\mathrm{C222}(1) \longleftrightarrow \ \mathrm{Ref}(\mathrm{ZF}) \ \longleftrightarrow \ \mathrm{Ref}(\mathrm{ZFC}). \tag{8}$$

The immediate calibration files are pinned to commit

`08200eb10fde2db408c69dffd840d6920a569b1f`.

Their complete byte counts and SHA-256 values were checked against the archive's ledger. The independent reviewer inspected the actual statements as well as the file identities. This investigation did not rebuild their full transitive Lean development.[2, 3]

The complete inherited archive is included as a frozen dependency, with SHA-256

`59f847872298260ee35e9d8e4cd0b52d2d78658ac523981fe060e81f40ca4a59`.

Later ordinary arithmetic reductions are also included with their original repository paths and hashes. These include the candidate-output factorial radius, signed comparison-base theorem, paired index guards, factorial lower-bound guard, and early first-output sign recovery. This package identifies and preserves that dependency chain rather than treating a finite Python replay as a substitute for its number theory or logical calibration.

4.4 Proof of the consistency endpoint

Proof of Theorem 1.1. The source arithmetic theorem identifies the existence of a valid numerical certificate with C222(1). The squared-output transformation in Proposition 2.2 and the quotient-radius transformation in Corollary 2.4 preserve that existence statement. The signed integer projections preserve it in both directions by exact polynomial restoration. The coefficient-height transformation preserves it by Theorem 3.2. Composing any of these paths with (8) gives

$$\exists \mathbf{x} \in \mathbb{Z}^d \ P(\mathbf{x}) = 0 \quad \longleftrightarrow \quad \exists p \ \mathrm{Proof}_{\mathrm{ZFC}}(p, \bot).$$

All fixed data, finite sequences, congruences, Pell recurrences and witness constructions are arithmetically definable. The ordinary arguments use induction, integer algebra, elementary number theory, and the identified source calibration, all formalizable in ZFC. Negating the displayed equivalence gives the stated no-zero equivalence with Con(ZFC). □

No consistency assumption is used in deriving the equivalence. If a finite syntactic contradiction exists, the completeness constructions produce the corresponding finite integer witnesses. If an integer zero exists, soundness produces a finite syntactic contradiction. Neither implication depends on our being able to exhibit such an object.

The statement "ZFC proves the equivalence" here has the ordinary mathematical meaning that the supplied proof and its named dependencies can be formalized in ZFC. It is distinct from a completed new proof-assistant theorem and from an emitted proof term in the repository's encoded object-language ZFC. The latter tasks remain unfinished, in agreement with the repository's formalization pause. This is a limitation on certification status, not a replacement of the no-zero claim by a bounded numerical test.

5 Verification, reproducibility, and proof status

5.1 What has been checked exactly

The finite checks complement the unbounded proofs. They establish that the supplied literal equations implement the stated transformations; they do not search for a ZFC contradiction or prove an infinite arithmetic theorem by testing examples.

The independent source reviewer imports no builder or optimizer. It checks the frozen original rows, exactly one changed body row, all 66 predicates and their private coordinates, 575 original prefix-row deduction identities, and 118 complete polynomial combinations of original rows. It independently reconstructs 605 usable prefix signs and confirms all 34 unsigned frame atoms. It binds the nine selected power calls to their actual exponents, retained index slacks and common base. It also checks all 19 division/remainder identities used to recover the omitted signs, including the radix identity required to bound the CRT modulus.

A separately authored projection verifier uses its own sparse-polynomial addition, multiplication and substitution routines. It replays every monic elimination over $\mathbb{Z}$, checks that no unconstrained coordinate was dropped, verifies the full retained-coordinate and retained-row partitions, and compares every surviving quadratic residual. It then expands the entire sum of squares and compares every coefficient of each of the four projected polynomials. Thus their variable counts, exact degrees, supports and coefficient heights are measured from complete literal equations.

For the height-two construction, a third independent checker verifies all 1,243 source-row pullback identities, 383 canonical old product definitions, 165 new products, 1,942 unit equations, all coefficient-collision conditions, and the entire 14,518-term polynomial. It verifies that all 3,564 coordinates occur, the constant term is one, and the degree is exactly four.

Focused corruption controls alter an inverse coefficient, a collected polynomial coefficient, the unit seed and recurrence, a product reconstruction, a source identity and a transformed row sign. They are rejected. The relevant source and coefficient checks also pass with Python optimization enabled, so their validation does not depend on disabled `assert` statements. Small numerical controls illustrate the signed-power inequalities and the essential base-one counterexample; the lemmas in the report supply the unbounded proofs.

5.2 Rebuilding the package

The supported numerical tools use Python 3 and its standard library. From the extracted package root, run

```
python3 verify_all.py
```

This authenticates the pinned source, checks both radius variants with the independent source reviewer, verifies all four projections and all their coefficients, and checks the preferred height-two equation.

Verification receipts are written under a temporary directory unless an explicit output location is requested.

For a deterministic rebuild of the two projected families, the builders support `-check`, which regenerates and compares their complete artifacts. The height-two builder can generate an independent output directory from its supplied source; the resulting normative files are byte-identical. Exact invocation examples are in the package README. Rebuilding the PDF additionally requires a standard LaTeX installation with the listed common packages.

The final appendix prints every residual of the principal 577-variable equation. The JSON coefficient lists remain authoritative for byte-level reproduction and include all four projected alternatives and the height-two polynomial. Each term `[c,[i,j,...]]` means $c\,x_i x_j \cdots$; an empty index list means the constant monomial. All indices are zero-based.

5.3 Certification levels are separate

Claim	Evidence delivered
The displayed files are fixed ordinary quartics	Complete residual and coefficient lists; exact expansion checks.
The projected zeros lift over all integers	Complete monic inverse certificates, checked in every row.
The omitted sign conditions are redundant after the radius change	Unbounded signed-power, magnitude and reconstruction proofs; independently checked source interfaces.
The coefficient-two equation has the same source zeros	Exact polynomial pullbacks and unique extension; full collision proof and coefficient collection.
The semantic source represents ZFC refutations	Identified inherited arithmetic proofs and C222/ZF/ZFC calibration, with pinned immediate source files.
The entire new endpoint has a proof-assistant proof	Not claimed. No new full Lean or encoded object-ZFC proof was completed.
The equations have no integer solutions	Not asserted independently of Con(ZFC).
Global simplicity or historical optimality	Not established.

The folder `quotient_frame/provenance/` contains the source map and later ordinary proofs. The inherited archive remains intact, including its notices. New scripts, source data and report are accompanied by a checksum inventory. The work is ready for incorporation as a separate research package; it does not change the repository's canonical displayed equation.

6 Interpretation and the next useful research steps

6.1 What made the saving possible

The useful structural distinction is between an arithmetic value and a certificate used to establish its sign. A proposed power output does not need to be positive before entering the power argument if its absolute size can be controlled independently. The quotient-radius construction obtains that control from the exact division rows, then spends a fixed margin in the exponent bound to exclude the unwanted Pell branch.

This suggests a systematic procedure for the direct compiler. Build a dependency graph of sign premises; identify terminal proposed values whose sign is used only to obtain a magnitude bound; express those magnitudes polynomially in earlier bounded data; and reserve enough margin below the shared exponent radius. Such a transformation must retain the actual index predicates and prove both branches of the

Pell orientation argument. The present nine deletions are a complete verified instance of that procedure, not a claim that every guard can be removed this way.

The coefficient-height construction addresses a different source of apparent complexity. Repeated linear occurrences, rather than individual arithmetic operations, cause many large coefficients in a sum of squares. Private units separate those occurrences while shared product wires keep the residuals quadratic. The complete zero-set bijection makes this a reusable compiler pass, with a transparent variable cost.

6.2 Tempting shortcuts that remain unproved or fail

The few-variable branch was examined but did not yield a sound new record in this investigation. In particular, shortening an auxiliary modulus can destroy the strict bound needed to recover a Pell index from a congruence. Removing a radicand factor can destroy the coprimality that turns a rational square product into separate integral squares. Neither proposed numerical saving is included as a result.

Several checks in the direct branch prevent similarly plausible mistakes. The signed-power lemma fails at base one. The quotient radius does not bound all old power outputs linearly. Old and new Pell tuples are not pointwise equal. A three-square guard cannot simply be deleted because the intended semantic value is positive: positivity must be recovered from the retained rows before interpreting that value. The new proofs make these dependencies explicit.

6.3 What would constitute a further substantial improvement

A further reduction should establish at least one new complete point in the measured variable/degree/height/support comparison. Three concrete possibilities remain: recover additional quotient signs with a checked index-bootstrap order; reduce the shared arithmetic source before it is compiled; or combine product sharing and coefficient control more efficiently while preserving a fully checked zero-set correspondence. The two retained exact-binomial high quotients need a different early-index argument, so deleting them is not licensed by this report.

The absolute fewest-variable problem is much harder. Classical universal-equation constructions and modern formalizations show that variable count and degree must be reported together.[4, 5] The present report deliberately does not rename a direct quartic improvement as a solution of that global optimization problem. Its completed output is a family of explicit equations with rigorous local improvements, a complete coefficient-two companion, and a reproducible path to the requested consistency equivalence.

References

[1] Vladimir Reshetnikov, *ProveIt*, inspected 24 October 2026. Frozen repository commit: `84e16e43092614bae262ca59651cb9d77e31a255`. Pinned DiophantineConsistency research tree. The baseline equations are in `research/factorial-radius-output-sign-profile/`; the current comparison is in `PROPOSAL_REVIEW_LATEST.md`.

[2] *ConZFC_407_888_Quartics_2026-10-24.zip*, preserved ProveIt incoming source archive. Pinned original archive. Included intact under `quotient_frame/provenance/`. It contains the direct calculus, CRT and arithmetic compilation proofs, source programs, notices, and immediate logical calibration files.

[3] ProveIt, the named logical bridge theorems:
`KEndpoint.c222_iff_zf_refutation`;
`ConstructibleChoice.zf_refutation_iff_zfc_refutation`.
Immediate source files pinned at commit
`08200eb10fde2db408c69dffd840d6920a569b1f`. Preserved in the preceding archive with complete source byte/hash ledger. Later ordinary arithmetic refinements are copied with their paths and hashes in `quotient_frame/provenance/INHERITED_SOURCE_MAP.json`.

[4] James P. Jones, "Universal Diophantine equation," *Journal of Symbolic Logic* 47 (1982), 549–571. doi:10.2307/2273588. Used only for historical context concerning universal-equation tradeoffs; the new direct constructions do not instantiate its polynomial.

[5] *Universal Pairs for Diophantine Equations*, Archive of Formal Proofs, 2025. Formalization entry. See also *A Formal Proof of Complexity Bounds on Diophantine Equations*, ITP 2025, LIPIcs ITP 2025, article 3.

A Complete residual list for the 577-variable equation

The unknowns are exactly $x_0, \dots, x_{576} \in \mathbb{Z}$. The single equation is

$$P_{577} = \sum_{j=0}^{362} r_j^2 = 0.$$

Every quadratic residual is printed below. No other symbols in this list stand for unknowns or unprinted coefficients. The row order is identical to `quotient_frame/support8/equation.json`. The fully expanded 7,326-term polynomial is supplied separately in the adjacent JSON file.

Row	Quadratic residual
$r_0 =$	$-x_4^2 + x_5$
$r_1 =$	$-x_5^2 + x_6$
$r_2 =$	$-x_6^2 + x_7$
$r_3 =$	$-x_7^2 + x_8$
$r_4 =$	$-x_8^2 + x_9$
$r_5 =$	$-x_9^2 + x_{10}$
$r_6 =$	$-x_{10}^2 + x_{11}$
$r_7 =$	$-x_{11}^2 + x_{12}$
$r_8 =$	$-x_{12}^2 + x_{13}$
$r_9 =$	$-x_{13}^2 + x_{14}$
$r_{10} =$	$-x_{14}^2 + x_{15}$
$r_{11} =$	$-x_{15}^2 + x_{16}$
$r_{12} =$	$-x_{16}^2 + x_{17}$
$r_{13} =$	$-x_{17}^2 + x_{18}$
$r_{14} =$	$-x_{18}^2 + x_{19}$
$r_{15} =$	$-x_{19}^2 + x_{20}$
$r_{16} =$	$-x_{20}^2 + x_{21}$
$r_{17} =$	$-x_{21}^2 + x_{22}$
$r_{18} =$	$-x_{22}^2 + x_{23}$
$r_{19} =$	$-x_{23}^2 + x_{24}$
$r_{20} =$	$-x_{24}^2 + x_{25}$
$r_{21} =$	$1 + x_3 - x_{25}^2$
$r_{22} =$	$-1 - x_0x_2 - 2x_2 + x_{28}$
$r_{23} =$	$-2 + x_1 - x_2x_{26} - x_3 - x_{26}$
$r_{24} =$	$-1 + x_1 - x_{27}x_{28}$
$r_{25} =$	$-9 - 6x_0 - x_0^2 - 2x_0x_1 - 2x_0x_2 - 2x_0x_3 - 6x_1 - x_1^2 - 2x_1x_2 - 2x_1x_3 - 6x_2 - x_2^2 - 2x_2x_3 - 6x_3$ $- x_3^2 + x_{38}$
$r_{26} =$	$-x_{38}^2 + x_{39}$
$r_{27} =$	$-x_{39}^2 + x_{40}$
$r_{28} =$	$-x_{40}^2 + x_{41}$
$r_{29} =$	$-x_{41}^2 + x_{42}$
$r_{30} =$	$-x_{42}^2 + x_{43}$
$r_{31} =$	$-x_{43}^2 + x_{44}$
$r_{32} =$	$-x_{44}^2 + x_{45}$
$r_{33} =$	$-x_{45}^2 + x_{46}$
$r_{34} =$	$-x_{46}^2 + x_{47}$

Row	Quadratic residual (continued)
$r_{35} =$	$-x_{47}^2 + x_{48}$
$r_{36} =$	$-x_{48}^2 + x_{49}$
$r_{37} =$	$-x_{49}^2 + x_{50}$
$r_{38} =$	$-x_{50}^2 + x_{51}$
$r_{39} =$	$-x_{51}^2 + x_{52}$
$r_{40} =$	$-x_{52}^2 + x_{53}$
$r_{41} =$	$-x_{53}^2 + x_{54}$
$r_{42} =$	$-x_{54}^2 + x_{55}$
$r_{43} =$	$-x_{55}^2 + x_{56}$
$r_{44} =$	$-x_{56}^2 + x_{57}$
$r_{45} =$	$-x_{57}^2 + x_{58}$
$r_{46} =$	$-x_{58}^2 + x_{59}$
$r_{47} =$	$-x_{59}^2 + x_{223} - x_{226}$
$r_{48} =$	$x_{61} - x_{223}^2 + 2x_{223}x_{226} - x_{226}^2$
$r_{49} =$	$-x_{60}x_{61} - x_{223} + x_{226} + x_{240} - x_{243}$
$r_{50} =$	$-1 - x_0x_{60} - 2x_{60} + x_{64}$
$r_{51} =$	$-1 - 2x_0 - x_0^2 + x_{68}$
$r_{52} =$	$-1 - x_0 - x_{67}x_{68} + x_{253} - x_{256}$
$r_{53} =$	$-x_{66}x_{67} + x_{72}$
$r_{54} =$	$1 - 2x_{60} - x_{60}x_{65} + x_{73}$
$r_{55} =$	$x_{71}x_{72} - x_{73}x_{75} - x_{74}$
$r_{56} =$	$-1 - x_{37}x_{60} - x_{60} + x_{74}$
$r_{57} =$	$-6 - 5x_{29} - x_{29}^2 - 2x_{29}x_{30} - 5x_{30} - x_{30}^2 + x_{76}$
$r_{58} =$	$-72 - 68x_{30} - 16x_{30}^2 - 16x_{30}x_{76} - 34x_{76} - 4x_{76}^2 + x_{77}$
$r_{59} =$	$-30 - 11x_{29} - x_{29}^2 - 4x_{29}x_{77} - 22x_{77} - 4x_{77}^2 + x_{78}$
$r_{60} =$	$-380 - 312x_{77} - 64x_{77}^2 - 32x_{77}x_{78} - 78x_{78} - 4x_{78}^2 + x_{79}$
$r_{61} =$	$-6 - 5x_{30} - x_{30}^2 - 2x_{30}x_{31} - 5x_{31} - x_{31}^2 + x_{80}$
$r_{62} =$	$-30 - 11x_{29} - x_{29}^2 - 8x_{29}x_{31} - 4x_{29}x_{80} - 44x_{31} - 16x_{31}^2 - 16x_{31}x_{80} - 22x_{80} - 4x_{80}^2 + x_{81}$
$r_{63} =$	$-6 - 5x_{29} - x_{29}^2 - 2x_{29}x_{31} - 5x_{31} - x_{31}^2 + x_{82}$
$r_{64} =$	$-156 - 100x_{30} - 16x_{30}^2 - 32x_{30}x_{31} - 16x_{30}x_{76} - 16x_{30}x_{82} - 100x_{31} - 16x_{31}^2 - 16x_{31}x_{76} - 16x_{31}x_{82}$ $- 50x_{76} - 4x_{76}^2 - 8x_{76}x_{82} - 50x_{82} - 4x_{82}^2 + x_{83}$
$r_{65} =$	$-1806 - 2720x_{31} - 1024x_{31}^2 - 512x_{31}x_{80} - 128x_{31}x_{81} - 512x_{31}x_{82} - 128x_{31}x_{83} - 680x_{80} - 64x_{80}^2$ $- 32x_{80}x_{81} - 128x_{80}x_{82} - 32x_{80}x_{83} - 170x_{81} - 4x_{81}^2 - 32x_{81}x_{82} - 8x_{81}x_{83} - 680x_{82} - 64x_{82}^2 - 32x_{82}x_{83}$ $- 170x_{83} - 4x_{83}^2 + x_{84}$
$r_{66} =$	$-110 - 84x_{29} - 16x_{29}^2 - 32x_{29}x_{30} - 16x_{29}x_{80} - 16x_{29}x_{82} - 84x_{30} - 16x_{30}^2 - 16x_{30}x_{80} - 16x_{30}x_{82} - 42x_{80}$ $- 4x_{80}^2 - 8x_{80}x_{82} - 42x_{82} - 4x_{82}^2 + x_{85}$
$r_{67} =$	$-756 - 1100x_{30} - 400x_{30}^2 - 80x_{30}x_{76} - 320x_{30}x_{80} - 80x_{30}x_{85} - 110x_{76} - 4x_{76}^2 - 32x_{76}x_{80} - 8x_{76}x_{85}$ $- 440x_{80} - 64x_{80}^2 - 32x_{80}x_{85} - 110x_{85} - 4x_{85}^2 + x_{86}$
$r_{68} =$	$-56 - 15x_{29} - x_{29}^2 - 8x_{29}x_{31} - 4x_{29}x_{80} - 60x_{31} - 16x_{31}^2 - 16x_{31}x_{80} - 30x_{80} - 4x_{80}^2 + x_{87}$
$r_{69} =$	$-56 - 15x_{30} - x_{30}^2 - 8x_{30}x_{31} - 4x_{30}x_{82} - 60x_{31} - 16x_{31}^2 - 16x_{31}x_{82} - 30x_{82} - 4x_{82}^2 + x_{88}$
$r_{70} =$	$-2756 - 3360x_{31} - 1024x_{31}^2 - 512x_{31}x_{80} - 512x_{31}x_{82} - 128x_{31}x_{87} - 128x_{31}x_{88} - 840x_{80} - 64x_{80}^2$ $- 128x_{80}x_{82} - 32x_{80}x_{87} - 32x_{80}x_{88} - 840x_{82} - 64x_{82}^2 - 32x_{82}x_{87} - 32x_{82}x_{88} - 210x_{87} - 4x_{87}^2 - 8x_{87}x_{88}$ $- 210x_{88} - 4x_{88}^2 + x_{89}$
$r_{71} =$	$-110 - 168x_{30} - 64x_{30}^2 - 32x_{30}x_{76} - 32x_{30}x_{80} - 42x_{76} - 4x_{76}^2 - 8x_{76}x_{80} - 42x_{80} - 4x_{80}^2 + x_{90}$
$r_{72} =$	$-756 - 220x_{29} - 16x_{29}^2 - 128x_{29}x_{30} - 64x_{29}x_{76} - 16x_{29}x_{82} - 16x_{29}x_{90} - 880x_{30} - 256x_{30}^2 - 256x_{30}x_{76}$ $- 64x_{30}x_{82} - 64x_{30}x_{90} - 440x_{76} - 64x_{76}^2 - 32x_{76}x_{82} - 32x_{76}x_{90} - 110x_{82} - 4x_{82}^2 - 8x_{82}x_{90} - 110x_{90}$ $- 4x_{90}^2 + x_{91}$
$r_{73} =$	$-56 - 60x_{30} - 16x_{30}^2 - 16x_{30}x_{76} - 30x_{76} - 4x_{76}^2 + x_{92}$

Row	Quadratic residual (continued)
$r_{74} =$	$-30-11x_{29}-x_{29}^2-4x_{29}x_{92}-22x_{92}-4x_{92}^2+x_{93}$
$r_{75} =$	$-380-312x_{92}-64x_{92}^2-32x_{92}x_{93}-78x_{93}-4x_{93}^2+x_{94}$
$r_{76} =$	$-30-11x_{30}-x_{30}^2-4x_{30}x_{94}-22x_{94}-4x_{94}^2+x_{95}$
$r_{77} =$	$-156-200x_{30}-64x_{30}^2-64x_{30}x_{76}-100x_{76}-16x_{76}^2+x_{96}$
$r_{78} =$	$-1406-1500x_{30}-400x_{30}^2-320x_{30}x_{76}-80x_{30}x_{80}-80x_{30}x_{96}-600x_{76}-64x_{76}^2-32x_{76}x_{80}-32x_{76}x_{96}$ $-150x_{80}-4x_{80}^2-8x_{80}x_{96}-150x_{96}-4x_{96}^2+x_{97}$
$r_{79} =$	$-16256-1020x_{29}-16x_{29}^2-512x_{29}x_{30}-256x_{29}x_{76}-16x_{29}x_{82}-64x_{29}x_{96}-16x_{29}x_{97}-16320x_{30}$ $-4096x_{30}^2-4096x_{30}x_{76}-256x_{30}x_{82}-1024x_{30}x_{96}-256x_{30}x_{97}-8160x_{76}-1024x_{76}^2-128x_{76}x_{82}$ $-512x_{76}x_{96}-128x_{76}x_{97}-510x_{82}-4x_{82}^2-32x_{82}x_{96}-8x_{82}x_{97}-2040x_{96}-64x_{96}^2-32x_{96}x_{97}-510x_{97}$ $-4x_{97}^2+x_{98}$
$r_{80} =$	$-6-5x_{31}-x_{31}^2-2x_{31}x_{32}-5x_{32}-x_{32}^2+x_{99}$
$r_{81} =$	$-30-11x_{29}-x_{29}^2-8x_{29}x_{32}-4x_{29}x_{99}-44x_{32}-16x_{32}^2-16x_{32}x_{99}-22x_{99}-4x_{99}^2+x_{100}$
$r_{82} =$	$-72-136x_{31}-64x_{31}^2-32x_{31}x_{80}-32x_{31}x_{82}-34x_{80}-4x_{80}^2-8x_{80}x_{82}-34x_{82}-4x_{82}^2+x_{101}$
$r_{83} =$	$-420-164x_{29}-16x_{29}^2-128x_{29}x_{31}-64x_{29}x_{80}-16x_{29}x_{100}-16x_{29}x_{101}-656x_{31}-256x_{31}^2-256x_{31}x_{80}$ $-64x_{31}x_{100}-64x_{31}x_{101}-328x_{80}-64x_{80}^2-32x_{80}x_{100}-32x_{80}x_{101}-82x_{100}-4x_{100}^2-8x_{100}x_{101}-82x_{101}$ $-4x_{101}^2+x_{102}$
$r_{84} =$	$-462-688x_{30}-256x_{30}^2-32x_{30}x_{31}-256x_{30}x_{80}-64x_{30}x_{85}-43x_{31}-x_{31}^2-16x_{31}x_{80}-4x_{31}x_{85}-344x_{80}$ $-64x_{80}^2-32x_{80}x_{85}-86x_{85}-4x_{85}^2+x_{103}$
$r_{85} =$	$-42-52x_{31}-16x_{31}^2-16x_{31}x_{80}-26x_{80}-4x_{80}^2+x_{104}$
$r_{86} =$	$-30-11x_{31}-x_{31}^2-4x_{31}x_{104}-22x_{104}-4x_{104}^2+x_{105}$
$r_{87} =$	$-306-35x_{30}-x_{30}^2-16x_{30}x_{104}-4x_{30}x_{105}-280x_{104}-64x_{104}^2-32x_{104}x_{105}-70x_{105}-4x_{105}^2+x_{106}$
$r_{88} =$	$-4556-4320x_{104}-1024x_{104}^2-512x_{104}x_{105}-128x_{104}x_{106}-1080x_{105}-64x_{105}^2-32x_{105}x_{106}-270x_{106}$ $-4x_{106}^2+x_{107}$
$r_{89} =$	$-30-11x_{31}-x_{31}^2-4x_{31}x_{107}-22x_{107}-4x_{107}^2+x_{108}$
$r_{90} =$	$-306-35x_{30}-x_{30}^2-16x_{30}x_{107}-4x_{30}x_{108}-280x_{107}-64x_{107}^2-32x_{107}x_{108}-70x_{108}-4x_{108}^2+x_{109}$
$r_{91} =$	$-22052-19008x_{30}-4096x_{30}^2-4096x_{30}x_{80}-1024x_{30}x_{85}-256x_{30}x_{103}-4096x_{30}x_{107}-1024x_{30}x_{108}$ $-256x_{30}x_{109}-9504x_{80}-1024x_{80}^2-512x_{80}x_{85}-128x_{80}x_{103}-2048x_{80}x_{107}-512x_{80}x_{108}-128x_{80}x_{109}$ $-2376x_{85}-64x_{85}^2-32x_{85}x_{103}-512x_{85}x_{107}-128x_{85}x_{108}-32x_{85}x_{109}-594x_{103}-4x_{103}^2-128x_{103}x_{107}$ $-32x_{103}x_{108}-8x_{103}x_{109}-9504x_{107}-1024x_{107}^2-512x_{107}x_{108}-128x_{107}x_{109}-2376x_{108}-64x_{108}^2$ $-32x_{108}x_{109}-594x_{109}-4x_{109}^2+x_{110}$
$r_{92} =$	$-70490-531x_{29}-x_{29}^2-256x_{29}x_{107}-64x_{29}x_{108}-16x_{29}x_{109}-4x_{29}x_{110}-67968x_{107}-16384x_{107}^2$ $-8192x_{107}x_{108}-2048x_{107}x_{109}-512x_{107}x_{110}-16992x_{108}-1024x_{108}^2-512x_{108}x_{109}-128x_{108}x_{110}-4248x_{109}$ $-64x_{109}^2-32x_{109}x_{110}-1062x_{110}-4x_{110}^2+x_{111}$
$r_{93} =$	$-1122540-1084928x_{107}-262144x_{107}^2-131072x_{107}x_{108}-32768x_{107}x_{109}-8192x_{107}x_{110}-2048x_{107}x_{111}$ $-271232x_{108}-16384x_{108}^2-8192x_{108}x_{109}-2048x_{108}x_{110}-512x_{108}x_{111}-67808x_{109}-1024x_{109}^2$ $-512x_{109}x_{110}-128x_{109}x_{111}-16952x_{110}-64x_{110}^2-32x_{110}x_{111}-4238x_{111}-4x_{111}^2+x_{112}$
$r_{94} =$	$-30-11x_{31}-x_{31}^2-4x_{31}x_{112}-22x_{112}-4x_{112}^2+x_{113}$
$r_{95} =$	$-306-35x_{29}-x_{29}^2-16x_{29}x_{112}-4x_{29}x_{113}-280x_{112}-64x_{112}^2-32x_{112}x_{113}-70x_{113}-4x_{113}^2+x_{114}$
$r_{96} =$	$-4556-4320x_{112}-1024x_{112}^2-512x_{112}x_{113}-128x_{112}x_{114}-1080x_{113}-64x_{113}^2-32x_{113}x_{114}-270x_{114}$ $-4x_{114}^2+x_{115}$
$r_{97} =$	$-90-76x_{31}-16x_{31}^2-16x_{31}x_{80}-16x_{31}x_{115}-38x_{80}-4x_{80}^2-8x_{80}x_{115}-38x_{115}-4x_{115}^2+x_{116}$
$r_{98} =$	$-306-35x_{30}-x_{30}^2-16x_{30}x_{115}-4x_{30}x_{116}-280x_{115}-64x_{115}^2-32x_{115}x_{116}-70x_{116}-4x_{116}^2+x_{117}$
$r_{99} =$	$-4556-4320x_{107}-1024x_{107}^2-512x_{107}x_{108}-128x_{107}x_{109}-1080x_{108}-64x_{108}^2-32x_{108}x_{109}-270x_{109}$ $-4x_{109}^2+x_{118}$
$r_{100} =$	$-4970-4512x_{115}-1024x_{115}^2-512x_{115}x_{116}-128x_{115}x_{117}-128x_{115}x_{118}-1128x_{116}-64x_{116}^2-32x_{116}x_{117}$ $-32x_{116}x_{118}-282x_{117}-4x_{117}^2-8x_{117}x_{118}-282x_{118}-4x_{118}^2+x_{119}$
$r_{101} =$	$-306-35x_{31}-x_{31}^2-16x_{31}x_{118}-4x_{31}x_{119}-280x_{118}-64x_{118}^2-32x_{118}x_{119}-70x_{119}-4x_{119}^2+x_{120}$
$r_{102} =$	$-42-52x_{30}-16x_{30}^2-8x_{30}x_{31}-16x_{30}x_{76}-13x_{31}-x_{31}^2-4x_{31}x_{76}-26x_{76}-4x_{76}^2+x_{121}$

Row	Quadratic residual (continued)
$r_{103} =$	$-756-1100x_{30}-400x_{30}^2-400x_{30}x_{76}-80x_{30}x_{121}-550x_{76}-100x_{76}^2-40x_{76}x_{121}-110x_{121}-4x_{121}^2+x_{122}$
$r_{104} =$	$-23870-19776x_{30}-4096x_{30}^2-4096x_{30}x_{76}-4096x_{30}x_{118}-1024x_{30}x_{119}-256x_{30}x_{120}-1024x_{30}x_{121}$ $-256x_{30}x_{122}-9888x_{76}-1024x_{76}^2-2048x_{76}x_{118}-512x_{76}x_{119}-128x_{76}x_{120}-512x_{76}x_{121}-128x_{76}x_{122}$ $-9888x_{118}-1024x_{118}^2-512x_{118}x_{119}-128x_{118}x_{120}-512x_{118}x_{121}-128x_{118}x_{122}-2472x_{119}-64x_{119}^2$ $-32x_{119}x_{120}-128x_{119}x_{121}-32x_{119}x_{122}-618x_{120}-4x_{120}^2-32x_{120}x_{121}-8x_{120}x_{122}-2472x_{121}-64x_{121}^2$ $-32x_{121}x_{122}-618x_{122}-4x_{122}^2+x_{123}$
$r_{105} =$	$-70490-67968x_{118}-16384x_{118}^2-8192x_{118}x_{119}-2048x_{118}x_{120}-512x_{118}x_{123}-16992x_{119}-1024x_{119}^2$ $-512x_{119}x_{120}-128x_{119}x_{123}-4248x_{120}-64x_{120}^2-32x_{120}x_{123}-1062x_{123}-4x_{123}^2+x_{124}$
$r_{106} =$	$-1-x_3-x_3x_{34}-x_{34}+x_{125}$
$r_{107} =$	$-x_{33}^2+2x_{33}x_{34}-x_{34}^2+x_{126}$
$r_{108} =$	$-1-x_{34}-2x_{79}x_{125}-4x_{125}+x_{127}$
$r_{109} =$	$-1-2x_{33}-x_{33}^2+2x_{33}x_{127}+2x_{127}-x_{127}^2+x_{128}$
$r_{110} =$	$-1-64x_{31}x_{125}-x_{34}-32x_{82}x_{125}-8x_{83}x_{125}-2x_{84}x_{125}-96x_{125}+x_{129}$
$r_{111} =$	$-1-2x_{33}-x_{33}^2+2x_{33}x_{129}+2x_{129}-x_{129}^2+x_{130}$
$r_{112} =$	$-1-64x_{30}x_{125}-x_{34}-32x_{80}x_{125}-8x_{85}x_{125}-2x_{86}x_{125}-80x_{125}+x_{131}$
$r_{113} =$	$-1-2x_{33}-x_{33}^2+2x_{33}x_{131}+2x_{131}-x_{131}^2+x_{132}$
$r_{114} =$	$-1-64x_{31}x_{125}-x_{34}-32x_{82}x_{125}-8x_{88}x_{125}-2x_{89}x_{125}-104x_{125}+x_{133}$
$r_{115} =$	$-1-2x_{33}-x_{33}^2+2x_{33}x_{133}+2x_{133}-x_{133}^2+x_{134}$
$r_{116} =$	$-1-64x_{30}x_{125}-x_{34}-32x_{76}x_{125}-8x_{90}x_{125}-2x_{91}x_{125}-80x_{125}+x_{135}$
$r_{117} =$	$-1-2x_{33}-x_{33}^2+2x_{33}x_{135}+2x_{135}-x_{135}^2+x_{136}$
$r_{118} =$	$-1-4x_{30}x_{125}-x_{34}-2x_{95}x_{125}-4x_{125}+x_{137}$
$r_{119} =$	$-1-2x_{33}-x_{33}^2+2x_{33}x_{137}+2x_{137}-x_{137}^2+x_{138}$
$r_{120} =$	$-1-256x_{30}x_{125}-x_{34}-128x_{76}x_{125}-32x_{96}x_{125}-8x_{97}x_{125}-2x_{98}x_{125}-480x_{125}+x_{139}$
$r_{121} =$	$-1-2x_{33}-x_{33}^2+2x_{33}x_{139}+2x_{139}-x_{139}^2+x_{140}$
$r_{122} =$	$-1-64x_{31}x_{125}-x_{34}-32x_{80}x_{125}-8x_{101}x_{125}-2x_{102}x_{125}-64x_{125}+x_{141}$
$r_{123} =$	$-1-2x_{33}-x_{33}^2+2x_{33}x_{141}+2x_{141}-x_{141}^2+x_{142}$
$r_{124} =$	$-x_{29}^2+2x_{29}x_{31}-x_{31}^2+x_{143}$
$r_{125} =$	$-x_{30}^2+2x_{30}x_{31}-x_{31}^2+x_{144}$
$r_{126} =$	$-x_{143}x_{144}+x_{145}$
$r_{127} =$	$-4-4x_{32}-x_{32}^2+x_{146}$
$r_{128} =$	$-x_{145}^2+2x_{145}x_{146}-x_{146}^2+x_{147}$
$r_{129} =$	$-1-x_{34}-2x_{124}x_{125}-4x_{125}+x_{148}$
$r_{130} =$	$-1-2x_{33}-x_{33}^2+2x_{33}x_{148}+2x_{148}-x_{148}^2+x_{149}$
$r_{131} =$	$-2-x_3-x_3x_{30}-x_{30}+x_{150}$
$r_{132} =$	$-x_{34}x_{150}-x_{150}+x_{151}$
$r_{133} =$	$-2-x_3-x_3x_{29}-x_{29}+x_{152}$
$r_{134} =$	$-5-4x_3-4x_3x_{30}-2x_3x_{76}-4x_{30}-2x_{76}+x_{153}$
$r_{135} =$	$-x_{33}x_{152}-x_{152}+x_{154}$
$r_{136} =$	$-x_{153}x_{154}+x_{155}$
$r_{137} =$	$-x_{151}^2+2x_{151}x_{155}-x_{155}^2+x_{156}$
$r_{138} =$	$-7-6x_3-4x_3x_{30}-2x_3x_{76}-4x_{30}-2x_{76}+x_{157}$
$r_{139} =$	$-x_{34}x_{157}-x_{157}+x_{158}$
$r_{140} =$	$-x_{33}x_{150}-x_{150}+x_{159}$
$r_{141} =$	$-x_{158}^2+2x_{158}x_{159}-x_{159}^2+x_{160}$
$r_{142} =$	$-x_{126}x_{128}+x_{161}$
$r_{143} =$	$-x_{130}x_{161}+x_{162}$
$r_{144} =$	$-x_{132}x_{162}+x_{163}$

Row	Quadratic residual (continued)
$r_{145} =$	$-x_{134}x_{163}+x_{164}$
$r_{146} =$	$-x_{136}x_{164}+x_{165}$
$r_{147} =$	$-x_{138}x_{165}+x_{166}$
$r_{148} =$	$-x_{140}x_{166}+x_{167}$
$r_{149} =$	$-x_{142}x_{167}+x_{168}$
$r_{150} =$	$-x_{147}x_{168}-x_{149}x_{168}+x_{169}$
$r_{151} =$	$-x_{156}x_{169}+x_{170}$
$r_{152} =$	$-1-x_{2}-x_{2}x_{37}+x_{171}$
$r_{153} =$	$-1-x_{33}-x_{35}x_{171}-x_{171}+x_{172}$
$r_{154} =$	$-2x_{1}^{2}+2x_{1}x_{172}+2x_{1}x_{173}+x_{74}x_{174}-x_{160}x_{170}-x_{172}^{2}-x_{173}^{2}$
$r_{155} =$	$-1-x_{2}-x_{2}x_{36}-x_{34}-x_{36}x_{171}-x_{171}+x_{173}$
$r_{156} =$	$-8-x_{29}-x_{30}-x_{31}-x_{32}-x_{33}-x_{34}-x_{35}-x_{36}+8x_{74}-x_{448}^{2}-x_{449}^{2}-x_{450}-x_{450}^{2}$
$r_{157} =$	$1+x_{62}-x_{63}x_{74}-x_{70}x_{74}-x_{74}+x_{177}$
$r_{158} =$	$-x_{62}-x_{62}x_{176}-x_{176}-x_{176}x_{177}+x_{178}$
$r_{159} =$	$-x_{62}x_{180}-x_{69}-x_{177}x_{180}-x_{180}+x_{181}$
$r_{160} =$	$-x_{62}x_{182}-x_{71}-x_{177}x_{182}-x_{182}+x_{183}$
$r_{161} =$	$-x_{74}x_{184}+x_{186}$
$r_{162} =$	$-x_{74}x_{190}+x_{192}$
$r_{163} =$	$-x_{74}x_{194}+x_{196}$
$r_{164} =$	$-x_{198}-x_{198}x_{200}+x_{201}$
$r_{165} =$	$-x_{201}^{2}+x_{202}$
$r_{166} =$	$1-x_{199}^{2}+x_{281}$
$r_{167} =$	$-1-2x_{198}-x_{198}^{2}+x_{203}$
$r_{168} =$	$x_{202}-x_{202}x_{203}+x_{281}$
$r_{169} =$	$-x_{205}^{2}+x_{208}$
$r_{170} =$	$-1+x_{204}^{2}-x_{208}x_{281}$
$r_{171} =$	$-1-4x_{205}-4x_{205}x_{206}+x_{209}$
$r_{172} =$	$-x_{199}-x_{204}x_{207}+x_{209}$
$r_{173} =$	$-x_{211}^{2}+x_{214}$
$r_{174} =$	$-1+x_{210}^{2}-x_{214}x_{281}$
$r_{175} =$	$-x_{213}^{2}+x_{215}$
$r_{176} =$	$-x_{209}^{2}+x_{216}$
$r_{177} =$	$-1+x_{212}^{2}+x_{215}-x_{215}x_{216}$
$r_{178} =$	$-x_{204}x_{217}-x_{210}+x_{212}$
$r_{179} =$	$-1-x_{0}-4x_{211}x_{218}+x_{213}$
$r_{180} =$	$-5-4x_{0}-x_{0}^{2}+2x_{0}x_{199}+4x_{199}-x_{219}$
$r_{181} =$	$-x_{199}x_{211}+x_{221}$
$r_{182} =$	$x_{0}x_{211}-x_{4}+x_{210}+2x_{211}-x_{219}x_{220}-x_{221}$
$r_{183} =$	$-x_{223}^{2}+x_{227}$
$r_{184} =$	$-1+x_{222}^{2}-x_{227}x_{281}$
$r_{185} =$	$-x_{225}^{2}+x_{228}$
$r_{186} =$	$-1-x_{216}x_{228}+x_{224}^{2}+x_{228}$
$r_{187} =$	$-x_{204}x_{229}-x_{222}+x_{224}$
$r_{188} =$	$-x_{223}-4x_{223}x_{230}+x_{225}+x_{226}$
$r_{189} =$	$-2+2x_{199}-2x_{199}x_{223}+2x_{199}x_{226}+2x_{199}x_{240}-2x_{199}x_{243}+2x_{223}-x_{223}^{2}+2x_{223}x_{226}+2x_{223}x_{240}$ $-2x_{223}x_{243}-2x_{226}-x_{226}^{2}-2x_{226}x_{240}+2x_{226}x_{243}-x_{231}-2x_{240}-x_{240}^{2}+2x_{240}x_{243}+2x_{243}-x_{243}^{2}$
$r_{190} =$	$x_{63}+x_{199}x_{223}-x_{222}-x_{223}+x_{223}^{2}-x_{223}x_{226}-x_{223}x_{240}+x_{223}x_{243}+x_{231}x_{232}$

Row	Quadratic residual (continued)
$r_{191} =$	$-1-x_{64}^2+2x_{64}x_{199}-x_{233}$
$r_{192} =$	$x_{64}x_{211}-x_{65}+x_{210}-x_{221}-x_{233}x_{234}$
$r_{193} =$	$-1-x_{60}^2+2x_{60}x_{199}-x_{235}$
$r_{194} =$	$x_{60}x_{211}-x_{66}+x_{210}-x_{221}-x_{235}x_{236}$
$r_{195} =$	$-1-x_0^2-2x_0x_{199}+2x_0x_{253}-2x_0x_{256}+2x_{199}x_{253}-2x_{199}x_{256}-x_{237}-x_{253}^2+2x_{253}x_{256}-x_{256}^2$
$r_{196} =$	$x_0x_{211}+x_{70}-x_{210}-x_{211}x_{253}+x_{211}x_{256}+x_{221}+x_{237}x_{238}$
$r_{197} =$	$-x_{240}^2+x_{244}$
$r_{198} =$	$-1+x_{239}^2-x_{244}x_{281}$
$r_{199} =$	$-x_{242}^2+x_{245}$
$r_{200} =$	$-1-x_{216}x_{245}+x_{241}^2+x_{245}$
$r_{201} =$	$-x_{204}x_{246}-x_{239}+x_{241}$
$r_{202} =$	$-x_{240}-4x_{240}x_{247}+x_{242}+x_{243}$
$r_{203} =$	$-5-4x_{62}-x_{62}^2-2x_{62}x_{177}+2x_{62}x_{199}-4x_{177}-x_{177}^2+2x_{177}x_{199}+4x_{199}-x_{248}$
$r_{204} =$	$-x_{62}x_{240}+x_{175}x_{178}-x_{177}x_{240}+x_{199}x_{240}-x_{239}-2x_{240}+x_{248}x_{249}+x_{454}^2+x_{455}^2+x_{456}+x_{456}^2$
$r_{205} =$	$-2-2x_{62}-x_{62}^2-2x_{62}x_{177}+2x_{62}x_{199}-2x_{177}-x_{177}^2+2x_{177}x_{199}+2x_{199}-x_{250}$
$r_{206} =$	$-x_{62}x_{223}+x_{175}-x_{177}x_{223}+x_{199}x_{223}-x_{222}-x_{223}+x_{250}x_{251}$
$r_{207} =$	$-x_{253}^2+x_{257}$
$r_{208} =$	$-1+x_{252}^2-x_{257}x_{281}$
$r_{209} =$	$-x_{255}^2+x_{258}$
$r_{210} =$	$-1-x_{216}x_{258}+x_{254}^2+x_{258}$
$r_{211} =$	$-x_{204}x_{259}-x_{252}+x_{254}$
$r_{212} =$	$-x_{253}-4x_{253}x_{260}+x_{255}+x_{256}$
$r_{213} =$	$-x_{62}x_{253}-x_{177}x_{253}+x_{179}x_{181}+x_{199}x_{253}+x_{248}x_{261}-x_{252}-2x_{253}+x_{463}^2+x_{464}^2+x_{465}+x_{465}^2$
$r_{214} =$	$-x_{62}x_{211}-x_{177}x_{211}+x_{179}-x_{210}-x_{211}+x_{221}+x_{250}x_{262}$
$r_{215} =$	$-x_{264}^2+x_{267}$
$r_{216} =$	$-1+x_{263}^2-x_{267}x_{281}$
$r_{217} =$	$-x_{266}^2+x_{268}$
$r_{218} =$	$-1-x_{216}x_{268}+x_{265}^2+x_{268}$
$r_{219} =$	$-x_{204}x_{269}-x_{263}+x_{265}$
$r_{220} =$	$-3-x_0-x_{65}-4x_{264}x_{270}+x_{266}$
$r_{221} =$	$-x_{62}x_{264}-x_{177}x_{264}+x_{179}x_{183}+x_{199}x_{264}+x_{248}x_{271}-x_{263}-2x_{264}+x_{475}^2+x_{476}^2+x_{477}+x_{477}^2$
$r_{222} =$	$-x_{273}^2+x_{276}$
$r_{223} =$	$-1+x_{272}^2-x_{276}x_{281}$
$r_{224} =$	$-x_{275}^2+x_{277}$
$r_{225} =$	$-1-x_{216}x_{277}+x_{274}^2+x_{277}$
$r_{226} =$	$-x_{204}x_{278}-x_{272}+x_{274}$
$r_{227} =$	$-1-x_{29}-4x_{273}x_{279}+x_{275}$
$r_{228} =$	$-x_{62}x_{273}-x_{177}x_{273}+x_{185}x_{186}+x_{199}x_{273}+x_{248}x_{280}-x_{272}-2x_{273}+x_{484}^2+x_{485}^2+x_{486}+x_{486}^2$
$r_{229} =$	$-x_{62}x_{210}-x_{62}x_{221}-x_{177}x_{210}-x_{177}x_{221}+x_{184}+x_{199}x_{221}-x_{210}-x_{211}x_{281}-x_{221}+x_{250}x_{282}$
$r_{230} =$	$-x_{284}^2+x_{287}$
$r_{231} =$	$-1-x_{281}x_{287}+x_{283}^2$
$r_{232} =$	$-x_{286}^2+x_{288}$
$r_{233} =$	$-1-x_{216}x_{288}+x_{285}^2+x_{288}$
$r_{234} =$	$-x_{204}x_{289}-x_{283}+x_{285}$
$r_{235} =$	$-1-x_{30}-4x_{284}x_{290}+x_{286}$
$r_{236} =$	$-x_{62}x_{284}-x_{177}x_{284}+x_{186}x_{187}+x_{199}x_{284}+x_{248}x_{291}-x_{283}-2x_{284}+x_{490}^2+x_{491}^2+x_{492}+x_{492}^2$
$r_{237} =$	$-x_{293}^2+x_{296}$

Row	Quadratic residual (continued)
$r_{238} =$	$-1-x_{281}x_{296}+x_{292}^2$
$r_{239} =$	$-x_{295}^2+x_{297}$
$r_{240} =$	$-1-x_{216}x_{297}+x_{294}^2+x_{297}$
$r_{241} =$	$-x_{204}x_{298}-x_{292}+x_{294}$
$r_{242} =$	$-1-x_{31}-4x_{293}x_{299}+x_{295}$
$r_{243} =$	$-x_{62}x_{293}-x_{177}x_{293}+x_{186}x_{188}+x_{199}x_{293}+x_{248}x_{300}-x_{292}-2x_{293}+x_{496}^2+x_{497}^2+x_{498}+x_{498}^2$
$r_{244} =$	$-x_{302}^2+x_{305}$
$r_{245} =$	$-1-x_{281}x_{305}+x_{301}^2$
$r_{246} =$	$-x_{304}^2+x_{306}$
$r_{247} =$	$-1-x_{216}x_{306}+x_{303}^2+x_{306}$
$r_{248} =$	$-x_{204}x_{307}-x_{301}+x_{303}$
$r_{249} =$	$-1-x_{32}-4x_{302}x_{308}+x_{304}$
$r_{250} =$	$-x_{62}x_{302}-x_{177}x_{302}+x_{186}x_{189}+x_{199}x_{302}+x_{248}x_{309}-x_{301}-2x_{302}+x_{502}^2+x_{503}^2+x_{504}+x_{504}^2$
$r_{251} =$	$-x_{311}^2+x_{314}$
$r_{252} =$	$-1-x_{281}x_{314}+x_{310}^2$
$r_{253} =$	$-x_{313}^2+x_{315}$
$r_{254} =$	$-1-x_{216}x_{315}+x_{312}^2+x_{315}$
$r_{255} =$	$-x_{204}x_{316}-x_{310}+x_{312}$
$r_{256} =$	$-1-x_{33}-4x_{311}x_{317}+x_{313}$
$r_{257} =$	$-x_{62}x_{311}-x_{177}x_{311}+x_{191}x_{192}+x_{199}x_{311}+x_{248}x_{318}-x_{310}-2x_{311}+x_{508}^2+x_{509}^2+x_{510}+x_{510}^2$
$r_{258} =$	$-x_{320}^2+x_{323}$
$r_{259} =$	$-1-x_{281}x_{323}+x_{319}^2$
$r_{260} =$	$-x_{322}^2+x_{324}$
$r_{261} =$	$-1-x_{216}x_{324}+x_{321}^2+x_{324}$
$r_{262} =$	$-x_{204}x_{325}-x_{319}+x_{321}$
$r_{263} =$	$-1-x_{2}-4x_{320}x_{326}+x_{322}$
$r_{264} =$	$-x_{62}x_{320}-x_{177}x_{320}+x_{190}+x_{199}x_{320}+x_{250}x_{327}-x_{319}-x_{320}$
$r_{265} =$	$-x_{329}^2+x_{332}$
$r_{266} =$	$-1-x_{281}x_{332}+x_{328}^2$
$r_{267} =$	$-x_{331}^2+x_{333}$
$r_{268} =$	$-1-x_{216}x_{333}+x_{330}^2+x_{333}$
$r_{269} =$	$-x_{204}x_{334}-x_{328}+x_{330}$
$r_{270} =$	$-1-x_{34}-4x_{329}x_{335}+x_{331}$
$r_{271} =$	$-x_{62}x_{329}-x_{177}x_{329}+x_{192}x_{193}+x_{199}x_{329}+x_{248}x_{336}-x_{328}-2x_{329}+x_{514}^2+x_{515}^2+x_{516}+x_{516}^2$
$r_{272} =$	$-x_{338}^2+x_{341}$
$r_{273} =$	$-1-x_{281}x_{341}+x_{337}^2$
$r_{274} =$	$-x_{340}^2+x_{342}$
$r_{275} =$	$-1-x_{216}x_{342}+x_{339}^2+x_{342}$
$r_{276} =$	$-x_{204}x_{343}-x_{337}+x_{339}$
$r_{277} =$	$-1-x_{35}-4x_{338}x_{344}+x_{340}$
$r_{278} =$	$-x_{62}x_{338}-x_{177}x_{338}+x_{195}x_{196}+x_{199}x_{338}+x_{248}x_{345}-x_{337}-2x_{338}+x_{520}^2+x_{521}^2+x_{522}+x_{522}^2$
$r_{279} =$	$-x_{347}^2+x_{350}$
$r_{280} =$	$-1-x_{281}x_{350}+x_{346}^2$
$r_{281} =$	$-x_{349}^2+x_{351}$
$r_{282} =$	$-1-x_{216}x_{351}+x_{348}^2+x_{351}$
$r_{283} =$	$-x_{204}x_{352}-x_{346}+x_{348}$
$r_{284} =$	$-1-x_{1}-4x_{347}x_{353}+x_{349}$

Row	Quadratic residual (continued)
$r_{285} =$	$-x_{62}x_{347} - x_{177}x_{347} + x_{194} + x_{199}x_{347} + x_{250}x_{354} - x_{346} - x_{347}$
$r_{286} =$	$-x_{356}^2 + x_{359}$
$r_{287} =$	$-1 - x_{281}x_{359} + x_{355}^2$
$r_{288} =$	$-x_{358}^2 + x_{360}$
$r_{289} =$	$-1 - x_{216}x_{360} + x_{357}^2 + x_{360}$
$r_{290} =$	$-x_{204}x_{361} - x_{355} + x_{357}$
$r_{291} =$	$-1 - x_{36} - 4x_{356}x_{362} + x_{358}$
$r_{292} =$	$-x_{62}x_{356} - x_{177}x_{356} + x_{196}x_{197} + x_{199}x_{356} + x_{248}x_{363} - x_{355} - 2x_{356} + x_{526}^2 + x_{527}^2 + x_{528} + x_{528}^2$
$r_{293} =$	$-x_{214}x_{227} + x_{364}$
$r_{294} =$	$-x_{244}x_{364} + x_{365}$
$r_{295} =$	$-x_{257}x_{365} + x_{366}$
$r_{296} =$	$-x_{267}x_{366} + x_{367}$
$r_{297} =$	$-x_{276}x_{367} + x_{368}$
$r_{298} =$	$-x_{287}x_{368} + x_{369}$
$r_{299} =$	$-x_{296}x_{369} + x_{370}$
$r_{300} =$	$-x_{305}x_{370} + x_{371}$
$r_{301} =$	$-x_{314}x_{371} + x_{372}$
$r_{302} =$	$-x_{323}x_{372} + x_{373}$
$r_{303} =$	$-x_{332}x_{373} + x_{374}$
$r_{304} =$	$-x_{341}x_{374} + x_{375}$
$r_{305} =$	$-x_{350}x_{375} + x_{376}$
$r_{306} =$	$-x_{359}x_{376} + x_{377}$
$r_{307} =$	$x_{205} - x_{377} - x_{377}x_{378}$
$r_{308} =$	$-25 - 2x_0 - x_1 - x_2 - x_4 - x_{29} - x_{30} - x_{31} - x_{32} - x_{33} - x_{34} - x_{35} - x_{36} - x_{60} - 2x_{62} - x_{63} - x_{64} - 2x_{65}$ $- x_{66} - x_{70} - x_{175} - x_{175}x_{178} - 2x_{177} - x_{179} - x_{179}x_{183} - x_{180}^2 - x_{184} - x_{185}^2 - x_{187}^2 - x_{188}^2 - x_{189}^2$ $- x_{190} - x_{191}^2 - x_{193}^2 - x_{194} - x_{195}^2 - x_{197}^2 + x_{198} - 2x_{240} + 2x_{243} - 2x_{253} + 2x_{256} - x_{454}^2 - x_{455}^2 - x_{456}$ $- x_{456}^2 - x_{475}^2 - x_{476}^2 - x_{477} - x_{477}^2$
$r_{309} =$	$x_0 - x_{379}^2 - x_{380}^2 - x_{381} - x_{381}^2$
$r_{310} =$	$x_3 - x_{382}^2 - x_{383}^2 - x_{384} - x_{384}^2$
$r_{311} =$	$x_4 - x_{385}^2 - x_{386}^2 - x_{387} - x_{387}^2$
$r_{312} =$	$x_{26} - x_{388}^2 - x_{389}^2 - x_{390} - x_{390}^2$
$r_{313} =$	$-2 + x_2 - x_3 - x_{391}^2 - x_{392}^2 - x_{393} - x_{393}^2$
$r_{314} =$	$x_{29} - x_{394}^2 - x_{395}^2 - x_{396} - x_{396}^2$
$r_{315} =$	$x_{30} - x_{397}^2 - x_{398}^2 - x_{399} - x_{399}^2$
$r_{316} =$	$x_{31} - x_{400}^2 - x_{401}^2 - x_{402} - x_{402}^2$
$r_{317} =$	$x_{32} - x_{403}^2 - x_{404}^2 - x_{405} - x_{405}^2$
$r_{318} =$	$x_{33} - x_{406}^2 - x_{407}^2 - x_{408} - x_{408}^2$
$r_{319} =$	$x_{34} - x_{409}^2 - x_{410}^2 - x_{411} - x_{411}^2$
$r_{320} =$	$x_{35} - x_{412}^2 - x_{413}^2 - x_{414} - x_{414}^2$
$r_{321} =$	$x_{36} - x_{415}^2 - x_{416}^2 - x_{417} - x_{417}^2$
$r_{322} =$	$x_{37} - x_{418}^2 - x_{419}^2 - x_{420} - x_{420}^2$
$r_{323} =$	$-1 - x_{60}x_{62} + x_{62} + x_{63} - x_{421}^2 - x_{422}^2 - x_{423} - x_{423}^2$
$r_{324} =$	$x_{60}x_{62} - x_{63} - x_{424}^2 - x_{425}^2 - x_{426} - x_{426}^2$
$r_{325} =$	$x_{65} - x_{427}^2 - x_{428}^2 - x_{429} - x_{429}^2$
$r_{326} =$	$x_{66} - x_{430}^2 - x_{431}^2 - x_{432} - x_{432}^2$
$r_{327} =$	$x_{70} - x_{433}^2 - x_{434}^2 - x_{435} - x_{435}^2$
$r_{328} =$	$-1 - x_{67}x_{69} + x_{69} + x_{70} - x_{436}^2 - x_{437}^2 - x_{438} - x_{438}^2$

Row	Quadratic residual (continued)
$r_{329} =$	$x_{67}x_{69} - x_{70} - x_{439}^2 - x_{440}^2 - x_{441} - x_{441}^2$
$r_{330} =$	$x_{71} - x_{442}^2 - x_{443}^2 - x_{444} - x_{444}^2$
$r_{331} =$	$-1 + x_{73} - x_{74} - x_{445}^2 - x_{446}^2 - x_{447} - x_{447}^2$
$r_{332} =$	$x_{176} - x_{451}^2 - x_{452}^2 - x_{453} - x_{453}^2$
$r_{333} =$	$-1 + x_{175} - x_{454}^2 - x_{455}^2 - x_{456} - x_{456}^2 - x_{457}^2 - x_{458}^2 - x_{459} - x_{459}^2$
$r_{334} =$	$x_{177} - x_{460}^2 - x_{461}^2 - x_{462} - x_{462}^2$
$r_{335} =$	$-1 + x_{179} - x_{463}^2 - x_{464}^2 - x_{465} - x_{465}^2 - x_{466}^2 - x_{467}^2 - x_{468} - x_{468}^2$
$r_{336} =$	$x_{62} - x_{69} + x_{177} - x_{469}^2 - x_{470}^2 - x_{471} - x_{471}^2$
$r_{337} =$	$x_{182} - x_{472}^2 - x_{473}^2 - x_{474} - x_{474}^2$
$r_{338} =$	$-1 + x_{179} - x_{475}^2 - x_{476}^2 - x_{477} - x_{477}^2 - x_{478}^2 - x_{479}^2 - x_{480} - x_{480}^2$
$r_{339} =$	$x_{62} - x_{71} + x_{177} - x_{481}^2 - x_{482}^2 - x_{483} - x_{483}^2$
$r_{340} =$	$-1 + x_{184} - x_{484}^2 - x_{485}^2 - x_{486} - x_{486}^2 - x_{487}^2 - x_{488}^2 - x_{489} - x_{489}^2$
$r_{341} =$	$-1 + x_{184} - x_{490}^2 - x_{491}^2 - x_{492} - x_{492}^2 - x_{493}^2 - x_{494}^2 - x_{495} - x_{495}^2$
$r_{342} =$	$-1 + x_{184} - x_{496}^2 - x_{497}^2 - x_{498} - x_{498}^2 - x_{499}^2 - x_{500}^2 - x_{501} - x_{501}^2$
$r_{343} =$	$-1 + x_{184} - x_{502}^2 - x_{503}^2 - x_{504} - x_{504}^2 - x_{505}^2 - x_{506}^2 - x_{507} - x_{507}^2$
$r_{344} =$	$-1 + x_{190} - x_{508}^2 - x_{509}^2 - x_{510} - x_{510}^2 - x_{511}^2 - x_{512}^2 - x_{513} - x_{513}^2$
$r_{345} =$	$-1 + x_{190} - x_{514}^2 - x_{515}^2 - x_{516} - x_{516}^2 - x_{517}^2 - x_{518}^2 - x_{519} - x_{519}^2$
$r_{346} =$	$-1 + x_{194} - x_{520}^2 - x_{521}^2 - x_{522} - x_{522}^2 - x_{523}^2 - x_{524}^2 - x_{525} - x_{525}^2$
$r_{347} =$	$-1 + x_{194} - x_{526}^2 - x_{527}^2 - x_{528} - x_{528}^2 - x_{529}^2 - x_{530}^2 - x_{531} - x_{531}^2$
$r_{348} =$	$x_{219} - x_{532}^2 - x_{533}^2 - x_{534} - x_{534}^2$
$r_{349} =$	$x_{256} - x_{535}^2 - x_{536}^2 - x_{537} - x_{537}^2$
$r_{350} =$	$-1 - x_{29} + x_{273} - x_{538}^2 - x_{539}^2 - x_{540} - x_{540}^2$
$r_{351} =$	$-1 - x_{30} + x_{284} - x_{541}^2 - x_{542}^2 - x_{543} - x_{543}^2$
$r_{352} =$	$-1 - x_{31} + x_{293} - x_{544}^2 - x_{545}^2 - x_{546} - x_{546}^2$
$r_{353} =$	$-1 - x_{32} + x_{302} - x_{547}^2 - x_{548}^2 - x_{549} - x_{549}^2$
$r_{354} =$	$-1 - x_{33} + x_{311} - x_{550}^2 - x_{551}^2 - x_{552} - x_{552}^2$
$r_{355} =$	$-1 - x_{2} + x_{320} - x_{553}^2 - x_{554}^2 - x_{555} - x_{555}^2$
$r_{356} =$	$-1 - x_{34} + x_{329} - x_{556}^2 - x_{557}^2 - x_{558} - x_{558}^2$
$r_{357} =$	$-1 - x_{35} + x_{338} - x_{559}^2 - x_{560}^2 - x_{561} - x_{561}^2$
$r_{358} =$	$-1 - x_{1} + x_{347} - x_{562}^2 - x_{563}^2 - x_{564} - x_{564}^2$
$r_{359} =$	$-1 - x_{36} + x_{356} - x_{565}^2 - x_{566}^2 - x_{567} - x_{567}^2$
$r_{360} =$	$-1 + x_{205} - x_{568}^2 - x_{569}^2 - x_{570} - x_{570}^2$
$r_{361} =$	$3 + 4x_0 + x_0^2 + x_0x_{65} - x_0x_{211} - x_0x_{264} + x_{65} - x_{65}x_{211} - 3x_{211} + x_{211}x_{264} - x_{264} - x_{571}^2 - x_{572}^2 - x_{573} - x_{573}^2$
$r_{362} =$	$-2 + x_{60} - x_{574}^2 - x_{575}^2 - x_{576} - x_{576}^2$